

Software Reverse Engineering Course

- Site: OnHexGroup.ir
- Youtube: @onhexgroup
- Telegram: onhex_ir
- Twitter: @onhexgroup
- Github: onhexgroup

Windows
(x86 / x64)

By: OnhexGroup

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Concept of Registers

- Registers are very fast memory locations that function like variables.

A=1



B=3

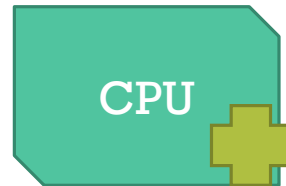
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Concept of Registers

A=1



B=3



A=?

B=?

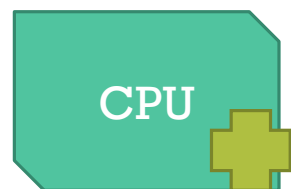
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Concept of Registers

A=1



B=3



A=?

B=?



3



1

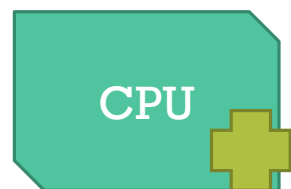
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Concept of Registers

A=1



B=3



A=?

B=?



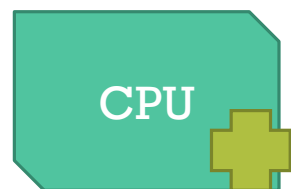
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Concept of Registers

A=1



B=3



A=?

B=?

0 1 2 3

1 3

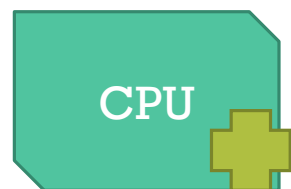
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Concept of Registers

A=1



B=3



A=?

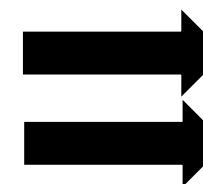
EAX

1

EBX

3

B=?



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Types of Basic Registers

- **General Purpose Registers**
- **Segment Registers**
- **Status and Control Registers**
- **Instruction Pointer Register**

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General Purpose Registers (GPR)

- **General Purpose Registers, also known as General Registers or GPRs**
- **Can be used in many situations. Can store both data and addresses**
- **Also have some specific, specialized roles**

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General Purpose Registers (GPR)

■ In 32-bit Systems

- **Maximum register size: 32 bits**
- **8 general-purpose registers:**
 - **EAX, EBX, ECX, EDX, ESI, EDI, EBP, ESP**

■ In 64-bit Systems

- **Maximum register size: 64 bits**
- **16 general-purpose registers:**
 - **RAX, RBX, RCX, RDX, RSI, RDI, RBP, RSP, R8–R15**
 - **Also referred to as R0–R15**

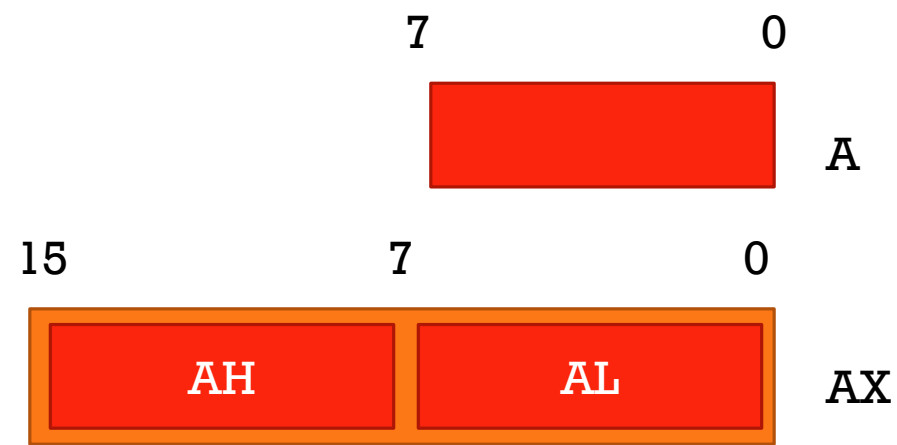
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Size of General Purpose Registers



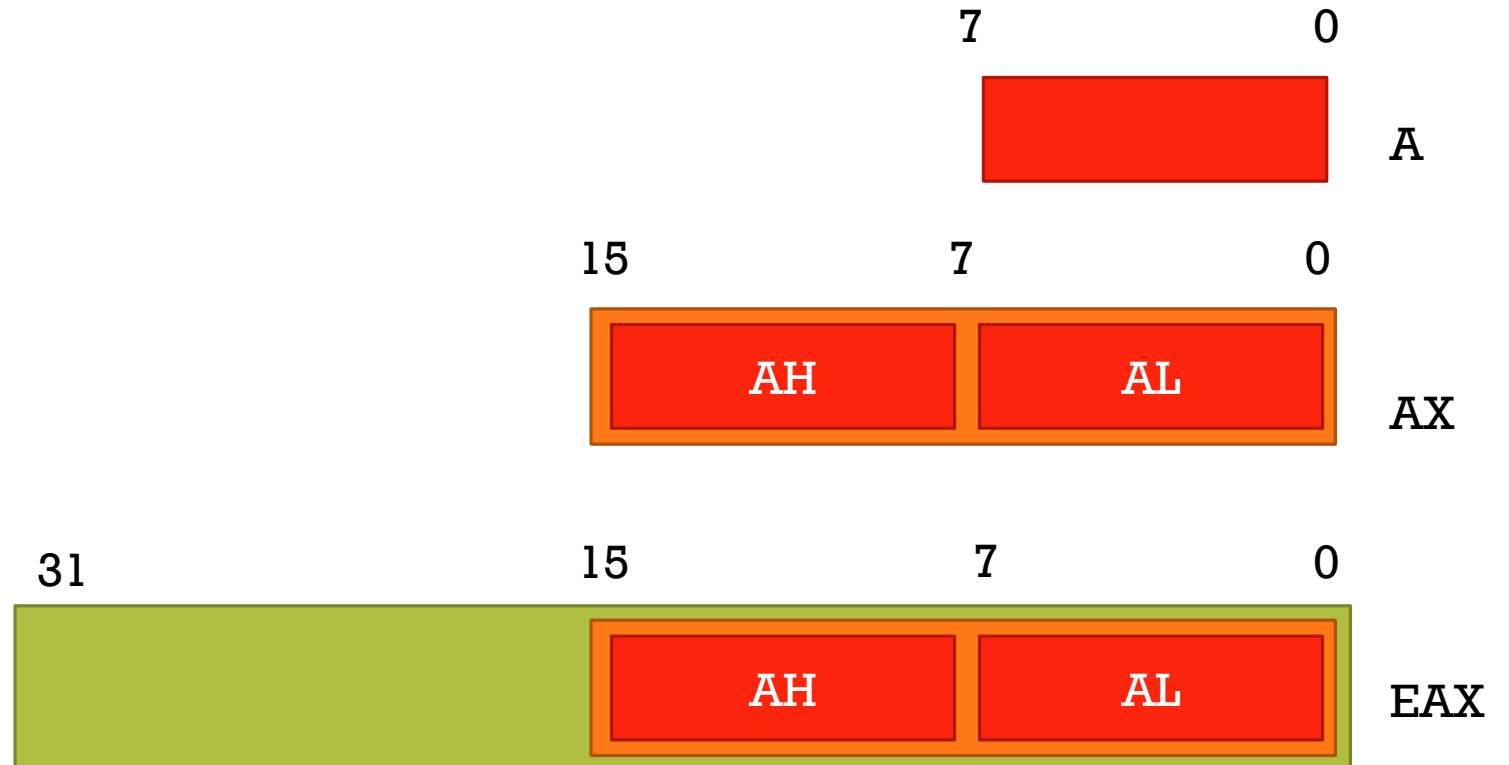
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Size of General Purpose Registers



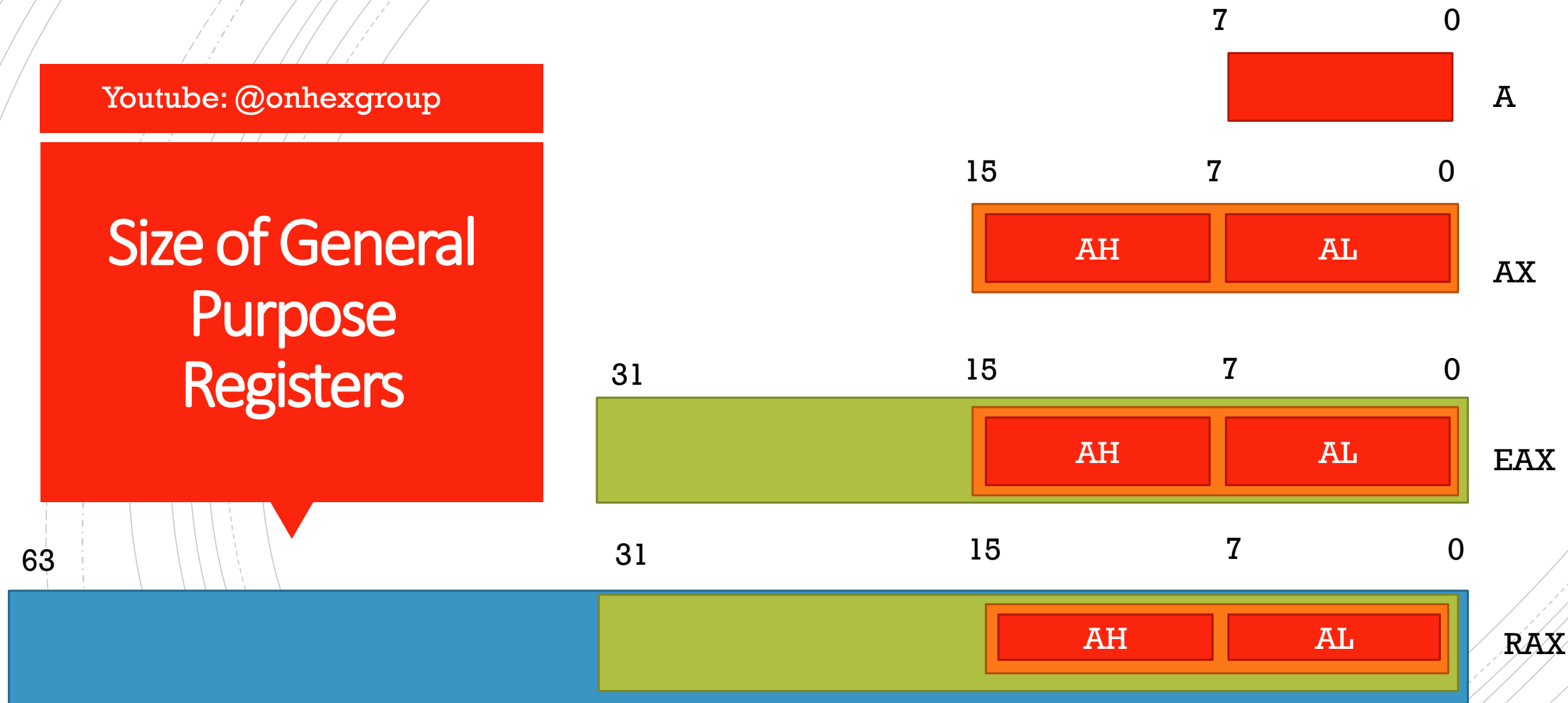
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Size of General Purpose Registers



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Size of General Purpose Registers



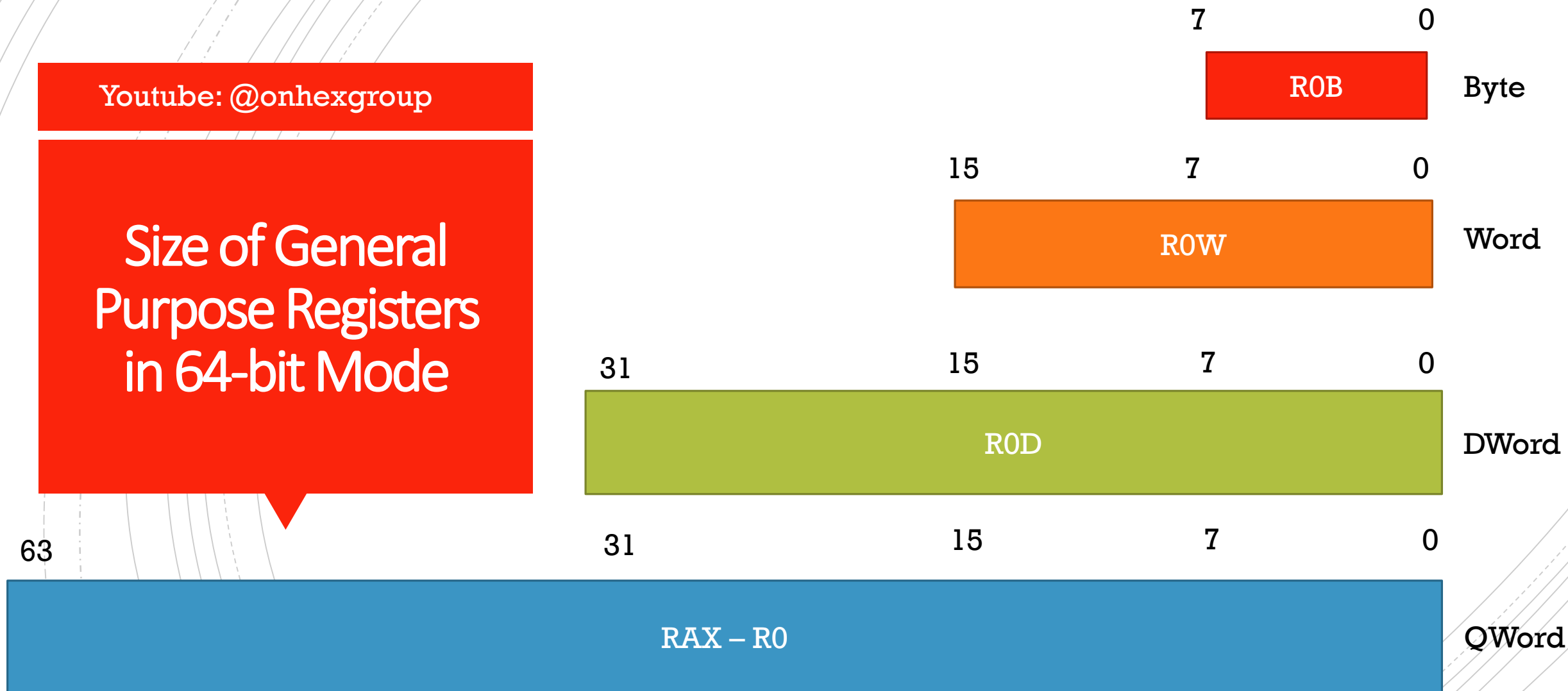
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Data Size of General Purpose Registers

- **AL / AH = 8 bits = 1 byte**
- **AX = 16 bits = 2 bytes = 1 word**
- **EAX = 32 bits = 4 bytes = 2 words = 1 DWORD**
- **RAX = 64 bits = 8 bytes = 4 words = 2 DWORDs = 1 QWORD**

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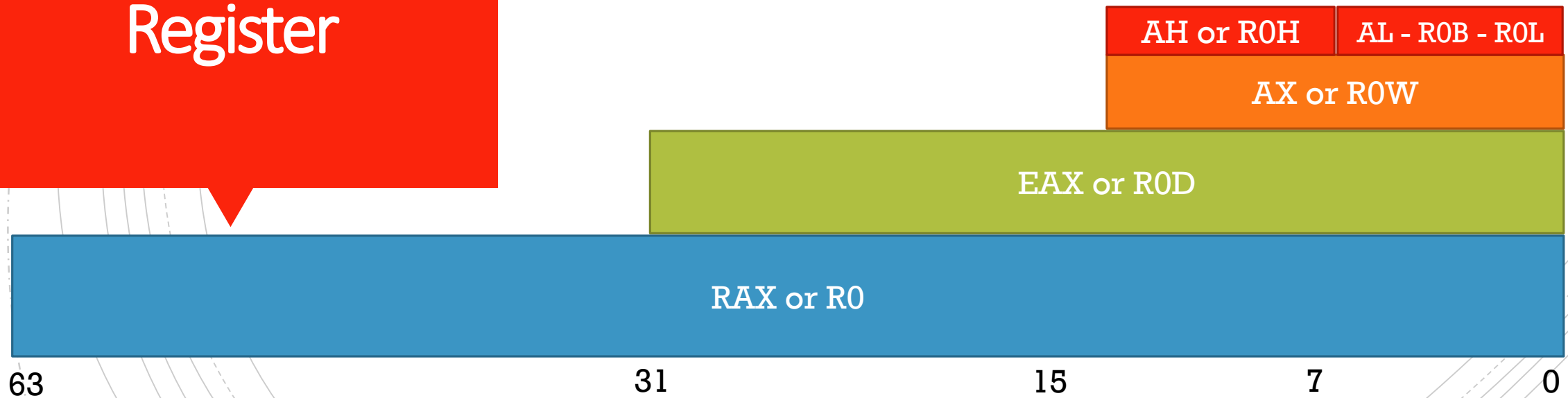
Size of General Purpose Registers in 64-bit Mode



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EAX / RAX / R0 Register

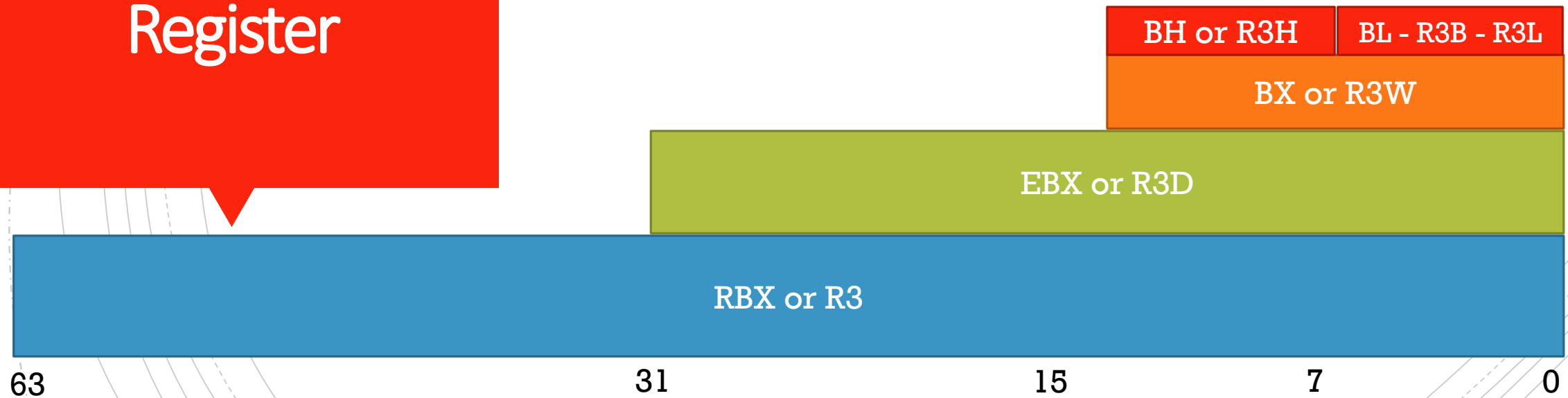
- **General-purpose register**
- **32-bit: EAX**
- **64-bit: RAX or R0**
- **Acts as the Accumulator**



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EBX / RBX / R3 Register

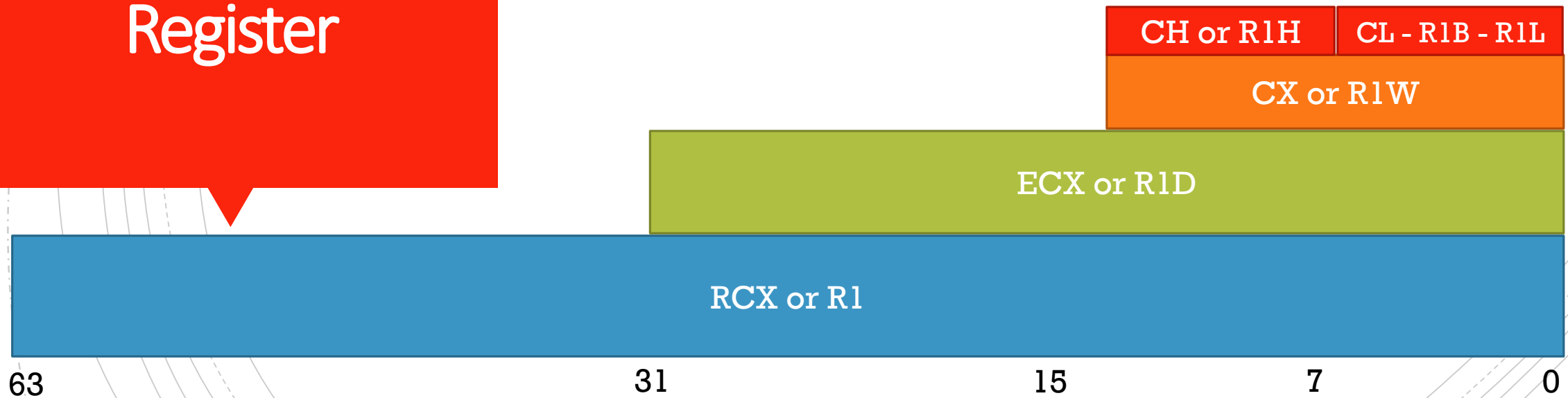
- General-purpose register
- 32-bit: **EBX**
- 64-bit: **RBX** or **R3**
- **Base Register**: stores the base address of the program



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ECX / RCX / R1 Register

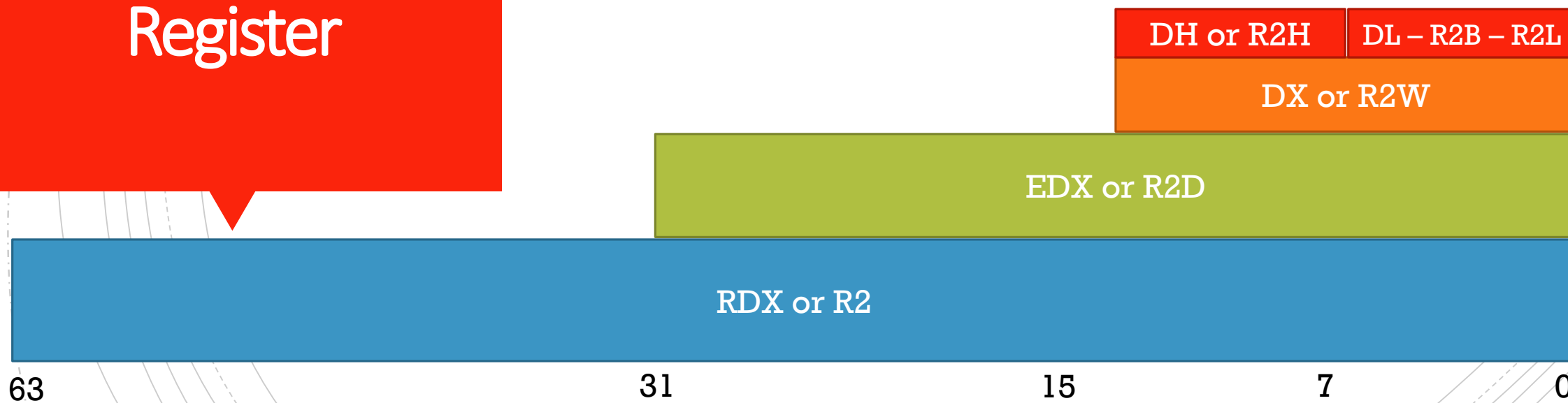
- General-purpose register
- 32-bit: **ECX**
- 64-bit: **RCX** or **R1**
- **Counter register**, used in loops and string operations



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EDX / RDX / R2 Register

- General-purpose register
- 32-bit: **EDX**
- 64-bit: **RDX** or **R2**
- **Data register**, commonly used in I/O operations



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ESP / RSP / R4
Register

- General-purpose register
- 32-bit: **ESP**
- 64-bit: **RSP** or **R4**
- **Stack Pointer**

Only Available in 64-bit Mode

SPL – R4B – R4L

SP or R4W

ESP or R4D

RSP or R4

63

31

15

7

0

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EBP / RBP / R5
Register

- General-purpose register
- 32-bit: **EBP**
- 64-bit: **RBP** or **R5**
- **Base Pointer**: points to stack frame data (end of stack frame)

Only Available in 64-bit Mode

BPL – R5B – R5L

BP or R5W

EBP or R5D

RBP or R5

63

31

15

7

0

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ESI / RSI / R6 Register

- General-purpose register
- 32-bit: **ESI**
- 64-bit: **RSI** or **R6**
- Data pointer, known as **Source Index** in string operations

Only Available in 64-bit Mode

SIL – R6B – R6L

SI or R6W

ESI or R6D

RSI or R6

63

31

15

7

0

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EDI / RDI / R7 Register

- General-purpose register
- 32-bit: **EDI**
- 64-bit: **RDI** or **R7**
- Data pointer, known as **Destination Index** in string operations

Only Available in 64-bit Mode

DIL – R7B – R7L

DI or R7W

EDI or R7D

RDI or R7

63

31

15

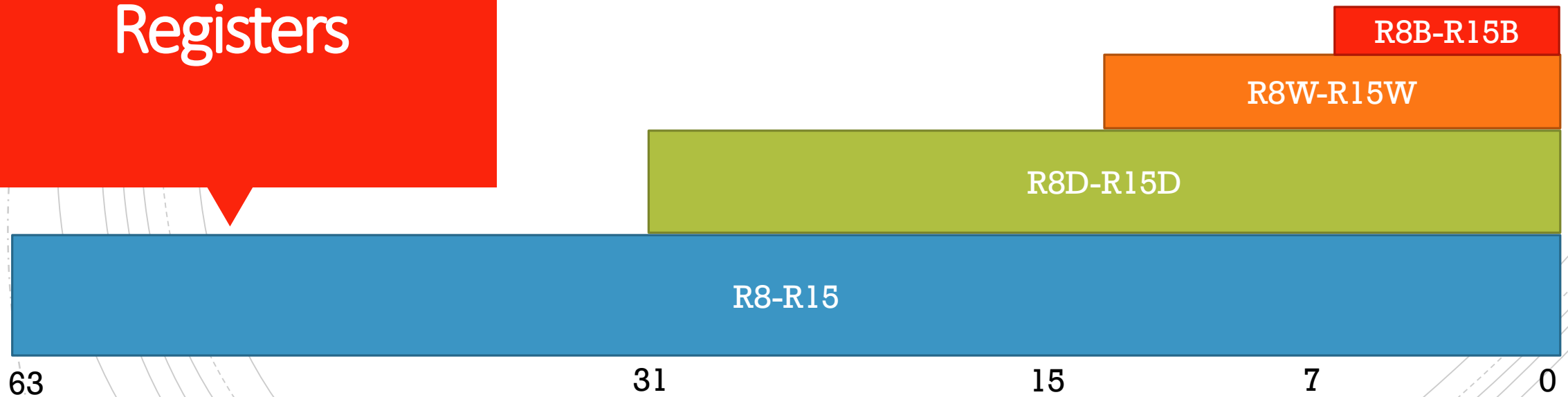
7

0

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R8–R15 Registers

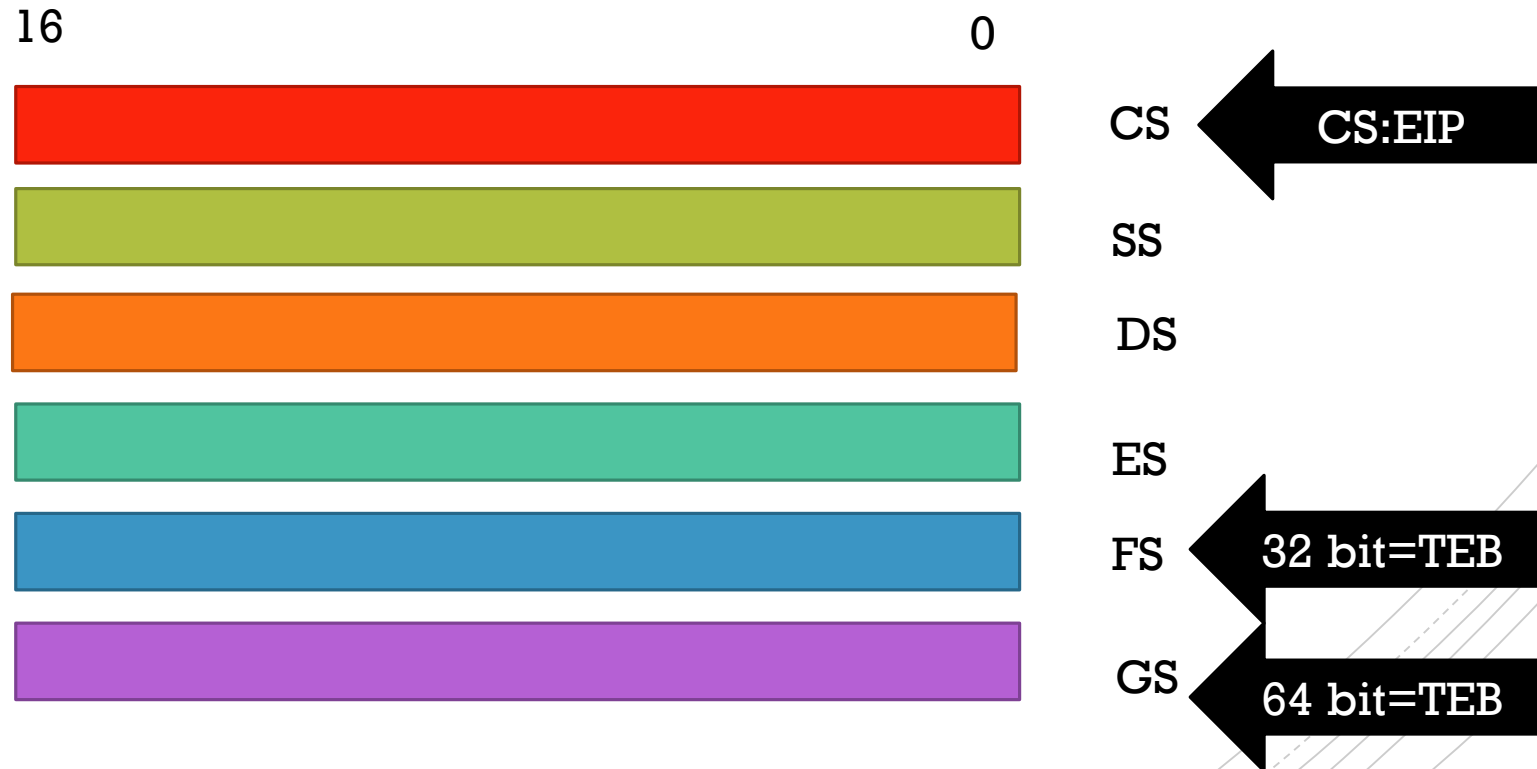
- New general-purpose registers available only in **64-bit architecture**
- Registers **R8 through R15**



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Segment Registers

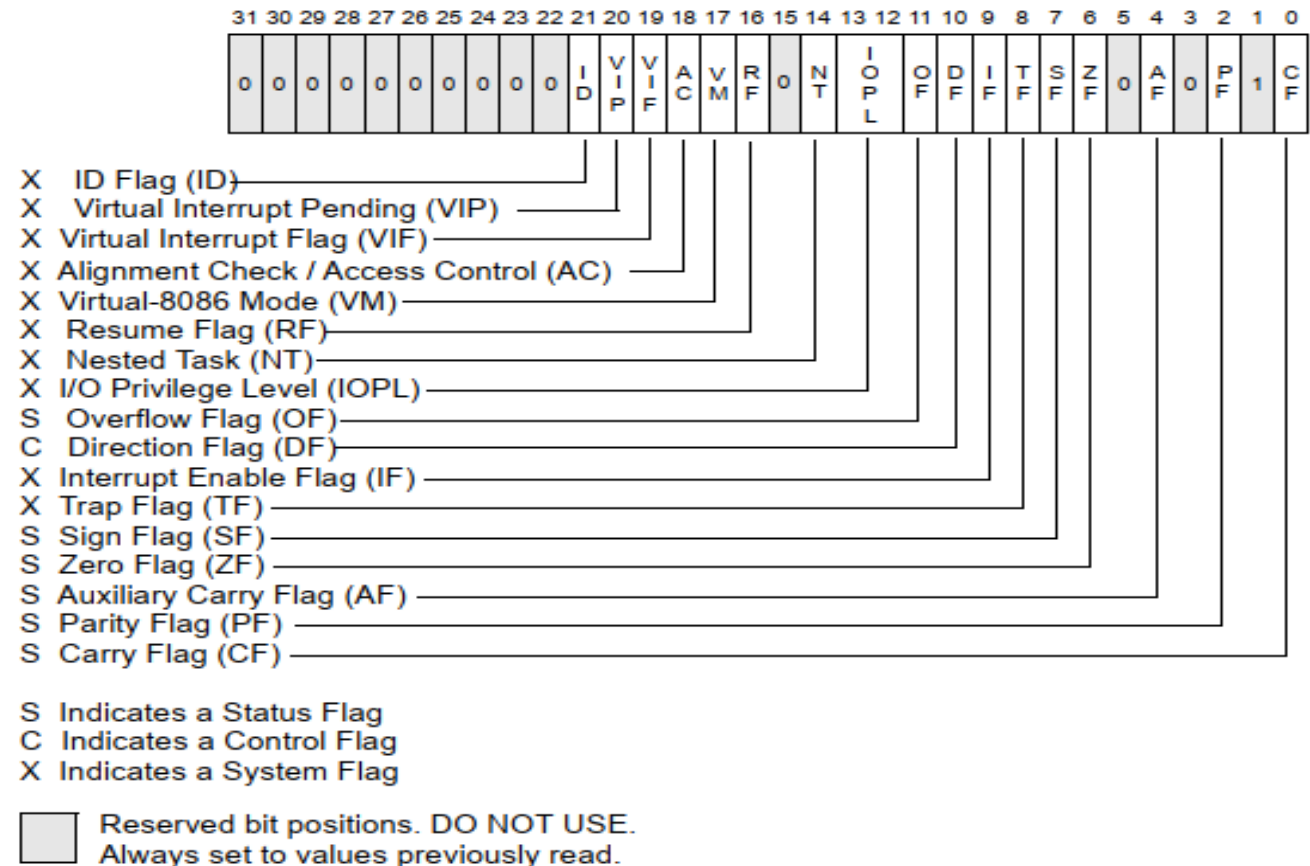
- A **segment** is a contiguous region of memory defined by a base address and a limit
- Purpose: memory division for protection and simpler control
- Segment registers point to these memory regions
- Rarely used in modern systems (**Flat Memory Model**)
- There are **6 segment registers**, each pointing to a specific memory segment (base address)



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Status and Control Registers EFLAGS / RFLAGS

- Each bit is called a **flag**
- Size in 32-bit systems: **32 bits**
- Size in 64-bit systems: **64 bits**
- Includes: System flags ,Control flags, Status flags



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Instruction Pointer Register EIP / RIP

- **Instruction Pointer Register**
- Holds the address (offset) of the next instruction to be executed
- Cannot be modified directly by the program
- Size:
 - 32-bit systems: **32 bits**
 - 64-bit systems: **64 bits**

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References

- <https://learn.microsoft.com/en-us/windows-hardware/drivers/debugger/x86-architecture>
- <https://learn.microsoft.com/en-us/windows-hardware/drivers/debugger/x64-architecture>
- https://en.wikibooks.org/wiki/X86_Assembly/X86_Architecture